

MATERIAL SAFETY DATA SHEET

CHEMICAL NAME:

PROPYLENE

HAZARD WARNING INFORMATION

HEALTH	FIRE	REACTIVE	OTHER	DEGREE OF HAZARD	COLOR CODING	OTHER CODES
1	4	1		0 = Minimum Hazard 1 = Slight Hazard 2 = Moderate Hazard 3 = Serious Hazard 4 = Severe Hazard	HEALTH = BLUE FIRE = RED REACTIVITY = YELLOW OTHER = WHITE	OX = Oxidizer ACID = Acid ALK = Alkali COR = Corrosive W = Use No Water

SECTION I - GENERAL INFORMATION

Characterization

Gas, Hydrocarbon

DOT Proper Shipping Name

Propylene

DOT Hazard Class and Label Requirements

Flammable Gas

DOT Identification Number

UN 1075 (compressed); UN 1077 (liquid)

Synonyms

Propene; methylethylene; methylethene.

RCRA Number

None

EPA Class

Not Applicable

Chemical Abstract Service (CAS) Number

115-07-1

DOT Emergency Guide Code

22

Chemical Formula

CH₃CH:CH₂

SECTION II - HAZARDOUS INGREDIENTS AND IDENTITY INFORMATION

Hazardous Components (specific identity)	OSHA Exposure Criteria	NIOSH Exposure Criteria	Immediately Dangerous to Life and Health (IDLH)	ACGIH Exposure Criteria
Propylene (derivation: Catalytic and thermal cracking of ethylene with zeolite catalyst, from naphtha).	PEL: Not Established	REL: Not Established	Not Determined	TLV: Not Established
	STEL: Not Established	STEL: Not Established		STEL: Not Established

SECTION III - PHYSICAL AND CHEMICAL CHARACTERISTICS

Boiling Point	-54°F (-48°C)	Specific Gravity (H ₂ O = 1)	0.51 (liquid)
Vapor Pressure (atmospheres)	10 at 68°F (20°C)	Molecular Weight	42
Vapor Density (Air = 1)	1.46 at 32°F (0°C)	Melting Point	-301°F (-185°C)

Solubility

Slightly soluble in water (0.01%). Soluble in alcohol and ether.

Appearance and Odor

Colorless gas with a slight odor, or a liquid when under pressure.

SECTION IV - FIRE AND EXPLOSION HAZARD DATA

Flash Point (method used)	-162°F (-108°C) closed cup	Explosive Limits in Air % by Volume	LEL: 2.0% UEL: 11.1%
NFPA Classification	Flammable Gas	Autoignition Temperature	927°F (497°C)

Extinguishing Media

Stop flow of gas to stop fire. If this is not possible, letting fire burn itself out may be only alternative.

Special Fire Fighting Procedures

Propylene is an extremely flammable, very dangerous fire and explosion hazard. Poisonous gases are evolved from fire. Wear a self-contained breathing apparatus (SCBA) with full facepiece. Ensure ALL sources of escaping gas are controlled. Monitor area long after fire is extinguished. Move containers from fire area if it can be done without risk. Use water spray to cool fire-exposed containers.

Unusual Fire and Explosion Hazards

Gas is explosive. Containers may explode in flame. Even if fire is extinguished, escaping gas can form explosive mixtures in air quickly and ignite without warning. Burns with a yellow, sooty flame.

SECTION V - REACTIVITY DATA				
Stability		Conditions to Avoid Propylene is normally stable when stored as a liquid in steel containers under its own pressure. Avoid contact with heat, sparks, flame, lighted tobacco products, and steam lines.		
Stable X	Unstable	Incompatibility (materials to avoid) Propylene explodes in contact with trifluoromethyl hypofluorite. It can react vigorously with oxidizing materials. Reacts with oxides of nitrogen to form an explosive product.		
Hazardous Polymerization		Conditions to Avoid Under normal conditions of temperature and pressure, hazardous polymerization of propylene is not expected to occur.		
May Occur	Will Not Occur X	Hazardous Decomposition or By-products If involved in fire, propylene can produce acrid and irritating smoke, fumes, and gases such as carbon monoxide and/or lower molecular weight hydrocarbons.		
SECTION VI - HEALTH HAZARD DATA				
Primary Route(s) of Entry:		Inhalation? X	Absorption (skin)?	Ingestion?
Health Hazards INHALATION: An asphyxiant gas. Initial symptoms are those of oxygen starvation. Feeling of “air hunger,” diminished mental alertness, headache, dullness, shortness of breath, and an impairment in muscle coordination. Continued exposures lead to faulty judgment, depression of senses, rapid fatigue, unconsciousness, convulsions, coma, and death. SKIN & EYES: Contact with the compressed gas or refrigerated liquid can cause frostbite and burns. INGESTION: Not likely since propylene is a gas at normal working (room) temperatures.				
Carcinogenicity Unknown Human Unknown Animal	NTP Listed? No	IARC Cancer Review Group? No	OSHA Regulated? No	Target Organs? Central nervous system
Medical Conditions Generally Aggravated by Exposure None reported.				
Emergency and First-aid Procedures <u>Eye contact:</u> Immediately flush with large amounts of warm (tepid) water for 15 minutes (minimum), occasionally lifting eyelids; seek medical attention. <u>Skin contact:</u> Remove all contaminated clothing. Immediately wash area with large amounts of warm (tepid) water. Do NOT rub or use air-blown heat on affected area. For <u>inhalation:</u> Remove the person from exposure. Provide respiratory assistance and CPR. Transfer promptly to medical facility. If <u>swallowed:</u> Not likely.				
SECTION VII - PRECAUTIONS FOR SAFE HANDLING AND USE				
Steps to be Taken in Case Material is Released or Spilled Remove all ignition sources. Ventilate area. Stop flow of gas. If the source of leak is a cylinder and cannot be stopped, remove to safe place in open air; repair leak or let cylinder empty. If gas has not yet ignited, use a water spray to knock down vapors and direct them from ignition source.				
Preferred Waste Disposal Method None specified in the references (incineration of a gas may be the only alternative).				
Precautions to be Taken in Handling and Storage Do not store with incompatible materials or in the presence of heat since violent reactions or explosions can occur. Store in tightly closed pressurized containers in a cool, dark, well-ventilated area away from sunlight. Keep away from ignition sources such as fire, sparks, and flame.				
Other Precautions and Warnings Bulk storage of propylene is not recommended. Containers may explode in fire or under conditions of extreme heat. Keep propylene out of confined areas or enclosed spaces (oxygen displacement hazard).				
SECTION VIII - CONTROL MEASURES AND PERSONAL PROTECTIVE EQUIPMENT				
Respiratory Protection (specify type) Propylene is a simple asphyxiant. Do NOT use cartridge-type respirators (ineffective for this type of gas). Use a powered supplied-air respirator or a self-contained breathing apparatus with full facepiece operated in pressure demand or other positive pressure mode.				
Ventilation Local exhaust or general mechanical systems recommended.				
Protective Gloves Insulated Neoprene or Polyurethane		Eye Protection Goggles and/or Face Mask		Other Protective Clothing Leather Apron
Work/Hygiene Practices Always wash hands thoroughly after using chemical; never bring food, drink, or smoking materials into vicinity of chemicals.				